Sales Analytics SQL: Practice Questions

Database: Anylises (schema dbo). Star schema with three tables:

- **dim_products**: product_key, product_id, product_number, product_name, category_id, category, subcategory, maintenance, cost, product_line, start_date
- **dim_customers**: customer_key, customer_id, customer_number, first_name, last_name, country, marital_status, gender, birthdate, create_date
- **fact_sales**: order_number, product_key, customer_key, order_date, shipping_date, due_date, sales_amount, quantity, price

1. Database Exploration

1. List all tables in the database using INFORMATION_SCHEMA.TABLES.
2. List all columns of each table with their data types using INFORMATION_SCHEMA.COLUMNS.
3. How many rows does each table have?
4. Which columns are shared keys between tables (the join columns)?
5. Are there any NULLs in key columns (product_key, customer_key)?
6. Are there duplicate keys in the dimension tables?

2. Dimension Exploration

1. Which countries do customers come from?
2. What are the distinct values of marital_status and gender?
3. What are all the distinct categories, subcategories and product lines?
4. Show the hierarchy of category, subcategory and product_name.
5. How many distinct products are in each category and subcategory?
6. Are there any products or customers with missing or "n/a" values in descriptive columns?

3. Date Exploration

1. What are the first and last order_date in fact_sales?
2. How many years and months of sales data are available?
3. Who is the youngest and the oldest customer (from birthdate), and what are their ages?
4. What is the earliest and latest customer create_date?

5. What is the earliest product start_date?
6. What is the average number of days between order_date and shipping_date? Between order_date and due_date?
7. Are there orders where shipping_date is later than due_date (late shipments)?

4. Measure Exploration (Big Numbers)

1. Total sales amount.
2. Total quantity sold.
3. Average selling price.
4. Total number of orders (distinct order_number).
5. Total number of products and total number of customers.
6. How many customers actually placed an order?
7. Build one report that shows all key metrics in a single result set (using UNION ALL: measure_name, measure_value).

5. Magnitude Analysis

1. Total customers by country.
2. Total customers by gender.
3. Total products by category.
4. Average cost per category.
5. Total revenue per category.
6. Total revenue per customer.
7. Distribution of sold items across countries.
8. Average sales amount per order by country.

6. Ranking Analysis

1. Top 5 products by revenue.
2. Bottom 5 worst-performing products by revenue.
3. Top 5 subcategories by revenue.
4. Top 10 customers by revenue.
5. 3 customers with the fewest orders.
6. Rank products by revenue within each category (RANK, DENSE_RANK, ROW_NUMBER).
7. Top 3 products in each category (using window functions, not TOP).

8. Top 1 customer in each country.

7. Change Over Time (Trends)

1. Total sales, customer count and quantity per year.
2. Total sales per month (across all years) to see seasonality.
3. Total sales per year-month using DATETRUNC / FORMAT / DATEPART.
4. Number of new customers per month (using create_date).
5. Running total of sales over time.
6. Moving average of price over time.
7. Month-over-month growth in sales.
8. Which month has the highest sales in each year?

8. Cumulative Analysis

1. Running total of sales by month.
2. Running total of sales per year, resetting each year.
3. Cumulative average price over time.
4. Year-to-date (YTD) sales for each month.
5. Cumulative number of new customers over time.

9. Performance Analysis

1. Yearly sales for each product compared to that product's average sales.
2. Yearly sales for each product compared to the previous year (LAG), with the difference and a flag: Increase / Decrease / No Change.
3. Month-over-month change in total sales.
4. Percentage change year-over-year per category.
5. Which products are consistently growing year after year?
6. Compare each customer's yearly spending vs their own average.
7. Compare each product's sales to the best-selling product (FIRST_VALUE or MAX OVER).

10. Part-to-Whole Analysis

1. What percentage of total sales does each category contribute?
2. Percentage of sales by country.

3. Percentage of sales by gender and marital status.
4. Percentage contribution of each product line.
5. Which subcategories contribute to the top 80% of sales (Pareto analysis)?
6. What share of orders does each country make up?
7. Each year's share of overall sales.

11. Data Segmentation

1. Segment products by cost range (Below 100, 100-500, 500-1000, Above 1000) and count products in each segment.
2. Segment customers by spending behavior and lifespan, then count customers in each group:
   - VIP: at least 12 months of history and spending over 5,000
   - Regular: at least 12 months of history and spending 5,000 or less
   - New: lifespan less than 12 months
3. Segment customers by age group (Under 20, 20-29, 30-39, 40-49, 50+).
4. Segment customers by order frequency (one-time, repeat, loyal).
5. Segment products by sales performance (High, Medium, Low performers).
6. Group customers by country and age group, and show total sales for each.

12. Bonus: Final Reports (Views)

1. Create the view **report_customers** with: customer name, age, age group, segment, total orders, total sales, total quantity, total products, lifespan (months), recency (months since last order), average order value, average monthly spend.
2. Create the view **report_products** with: product name, category, subcategory, cost, total sales, total orders, total customers, lifespan, recency, segment (High/Mid/Low performer), average order revenue, average monthly revenue.